IMAGES OF THE LIMIT INSIDE A LOCALLY CONSTRUCTIBLE SET ARE EVENTUAL

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Constructible sets as images. Algebra, Lemma algebra-lemma-constructible-is-image
- Limit fibre product empty. Limits, Lemma lemma-limit-fibre-product-empty (same chapter)

Source reference. Original reference 086P: sources/stacks-chow/tags.txt, source lines 10399–10399. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01YX: sources/stacks-chow/tags.txt, source lines 2572–2572. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 054C: sources/stacks-chow/tags.txt, source lines 6635–6635. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00F8: sources/stacks-chow/tags.txt, source lines 562–562. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05F3: sources/stacks-chow/tags.txt, source lines 7012–7012. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 086P. Suppose we are given an i and a locally constructible subset $E \subset S_i$ such that $f_i(S) \subset E$. Then $f_{i'i}(S_{i'}) \subset E$ for all sufficiently large i'.*

Proof. Writing S_i as a finite union of open affine subschemes reduces the question to the case that S_i is affine and E is constructible, see Lemma 01YX and Properties, Lemma 054C. In this case the complement $S_i \setminus E$ is constructible too. Hence there exists an affine scheme T and a morphism $T \to S_i$ whose image is $S_i \setminus E$, see Algebra, Lemma 00F8. By Lemma 05F3 we see that $T \times_{S_i} S_{i'}$ is empty for all sufficiently large i', and hence $f_{i'i}(S_{i'}) \subset E$ for all sufficiently large i'. □

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